INTERIM AMENDMENT

TO

FEDERAL SPECIFICATION

VARNISH, OIL (LOW SHEEN, BRUSH OR SPRAY APPLICATION)

This interim amendment was developed by the General Services Administration, Office of Personal Property, Washington, DC 20406, based on currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this interim amendment as a valid exception to TT-V-85C, dated April 5, 1976 and Amendment - 1, dated June 23, 1977.

PAGE 2

Paragraph 2.2, Under "American Society for Testing and Materials (ASTM) Standards" delete "D 1306 - Phthalic Anhydride Content of Alkyd Resins and Esters Containing other Dibasic Acids Gravimetric" and "D 1475 - Density of Paint, Varnish, Lacquer and Related Products." Add "ASTM D 523 - Specular Gloss, ASTM D 1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes, and ASTM D 1544 - Color of Transparent Liquids Gardner Color Scale)."

Delete "Air Pollution Control District, County of Los Angeles:

Rule 66 - Organic Solvents and application for copies paragraph and substitute the following:

"Southern California Air Pollution Control District:

Rules 442, 443, and 102 Regulating the Use of Organic Solvents.

(Copies may be obtained from the State of California Air Resources Board, 1102 Q Street, P.O. Box 2815, Sacramento, CA 95828.)"

Paragraph 3.1; delete in its entirety and substitute the following: "3.1 Material. The varnish as received shall be clear, free from gel structures and sediment, and ready for use. The solvents shall comply with Rules 442, 443 and 102 Regulating the use of organic solvents."

Paragraph 3.3.2; delete in its entirety and substitute the following: "3.3.2 Color. When tested as specified in ASTM D 1544 the varnish shall be not darker than Gardner Liquid Color Standard No. 14."

Table I, Quantitative requirements - Under "characteristics" delete "Weight per gallon (pounds)" and "Phthalic anhydride (percent by weight of vehicle). " Delete the maximum requirement of "40" under gloss and substitute "50." Delete the maximum requirement of "40" under Total solids. Add; "Lead (percent nonvolatile);" under minimum "---", and under maximum "0.06".

PAGE 4

Table III. Index - Under "characteristics" delete "Weight per gallon" and "phthalic anhydride." Under the "ASTM" column add "D 1308" for water resistance, "D 1544 for color, and "D 523" for gloss. Under "Fed. Test Method Std. No. 141" delete "6101" for gloss and add "4.4.10" under "paragraph reference." Under "Paragraph reference "delete "4.4.2" for color, under "characteristics" add reference; "Table I", under Fed. Test Method Std. No. 141, "---"; under ASTM "---" and under paragraph reference "4.4.9".

Delete paragraph 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

Paragraph 4.4.7 - Delete in its entirety, and substitute the following:

"4.4.7 Water resistance. Draw down the varnish, using a doctor blade to a wet film thickness of 3.0 ± 0.1 mils on a solvent-cleaned tin panel, manufacturer's standard gage No. 31. Allow to dry 48 hours at standard conditions. In accordance with ASTM D 1308, spot the varnish with boiling distilled water, cover 30 minutes, rinse with distilled water, allow 3 hours to recover, and evaluate for compliance with 3.3.7."

Add new paragraphs:

4.4.9 Lead content.

4.4.9.1 Sample preparation. Using a 0.006-inch film applicator and a mechanical applicator plate, duplicate drawdowns for each sample of well-mixed paint shall be made on a standard paint penetration chart and dried for 24 hours. The drawdown shall be at least 10 inches long on the sealed portion of the penetration chart. The drawdown shall be cut into discs of appropriate size to fit the sample holder of a fluorescence X-ray spectrometer.

4.4.9.2 Procedure. Lead content shall be determined using an X-ray fluorescence spectrometer capable of determining lead content at a minimum level of 0.03 percent by weight of the total nonvolatile. The settings for a wavelength dispersive fluorescence spectrometer shall be as follows:(1)

Element	Analytical Line	Angle	Crystal	Detection	Collimeter	X-ray tube (MO)
Pb	L	33.93	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd I)		33.00	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd II)		35.50	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Mo	K	20.33	LiF(200)	Flow S.C.	Fine	60KV 45Ma

Pulse height selection shall be used in all measurements and counting time shall be 100 seconds. Place the sample disc in the wavelength dispersive unit. Measure the count rates of lead, lead background, and the Molybdenum Compton scattered background from the X-ray tube.

4.4.9.3 Calculation.

$$R = \frac{I_{Pb} - \frac{I_{Pb\ (Background\ I)} + I_{Pb\ (Background\ II)}}{2}}{I_{Mo}}$$

where I equals gross intensity. These results shall be compared to those obtained with a 0.06 percent lead standard made up from the same type of paint sample and evaluated for compliance with table I.

(1) Energy dispersive fluorescence spectrometers shall be set up according to the manufacturer's manual.

4.4.10 Gloss. On a sealed white test chart, draw down a film of the varnish using a doctor blade to a wet film thickness of 3.0 ± 0.1 mils. Allow to dry 48 hours at standard conditions in a dustfree environment. Determine gloss in accordance with ASTM D 523 and evaluate for compliance with 3.2."

PREPARING ACTIVITY:

GSA - OPP

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.